PROGRAMMING FUNDAMENTALS

LABORATORY ASSIGNMENT — 05

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Section: D

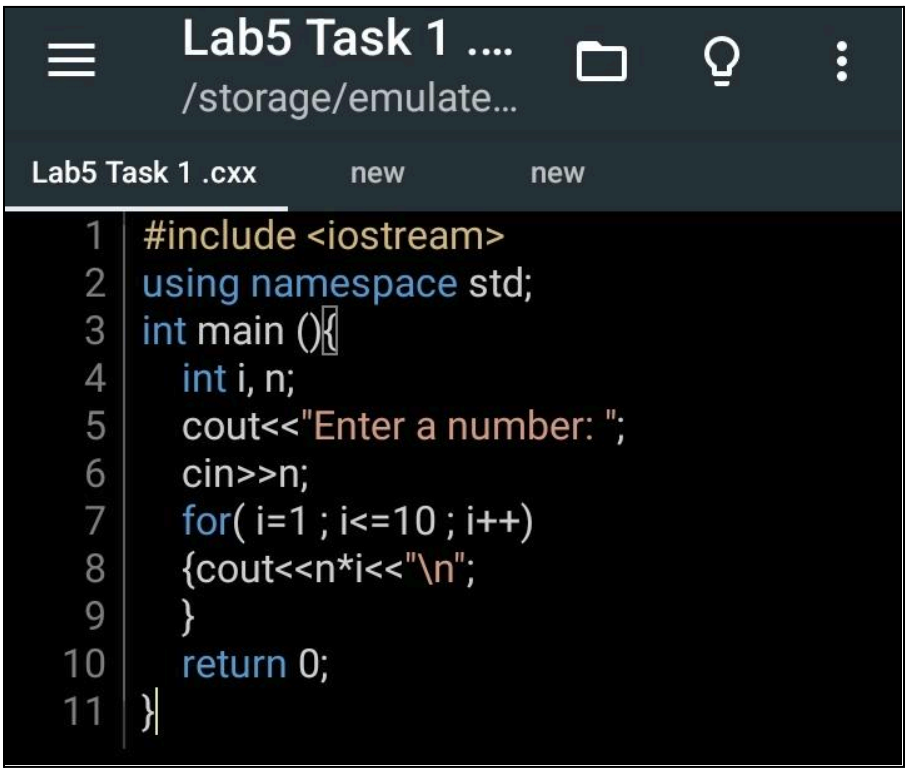
Department: BS Computer Science

Subject: Programming Fundamentals

Lab: PF Lab 05

Task – 01: (Print Table of any Number.)

Program:



Output:



TASK - 01
(Print Table Of Any Number)

Program:

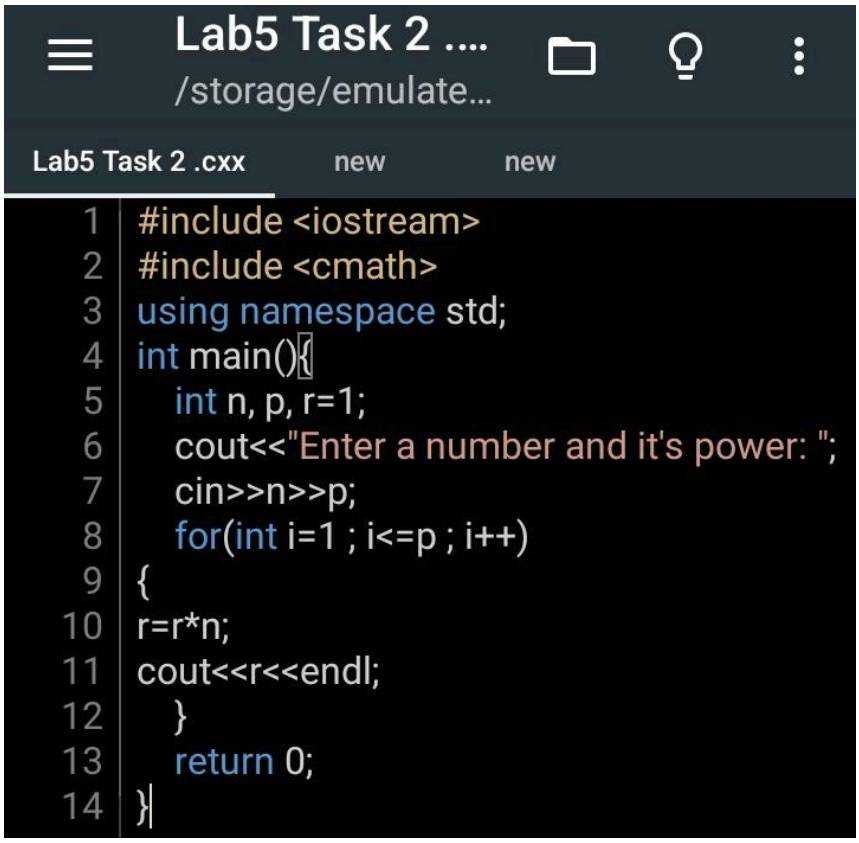
```
#include <iostream>
using namespace std;
int main(){
  int i, n;
  cout<<"Enter a number: ";
  cin>>n;
  for(i=1, i<=10; i++)
   {cout<<n*i<<"\n";
  }
  return 0;
  }
```

Output

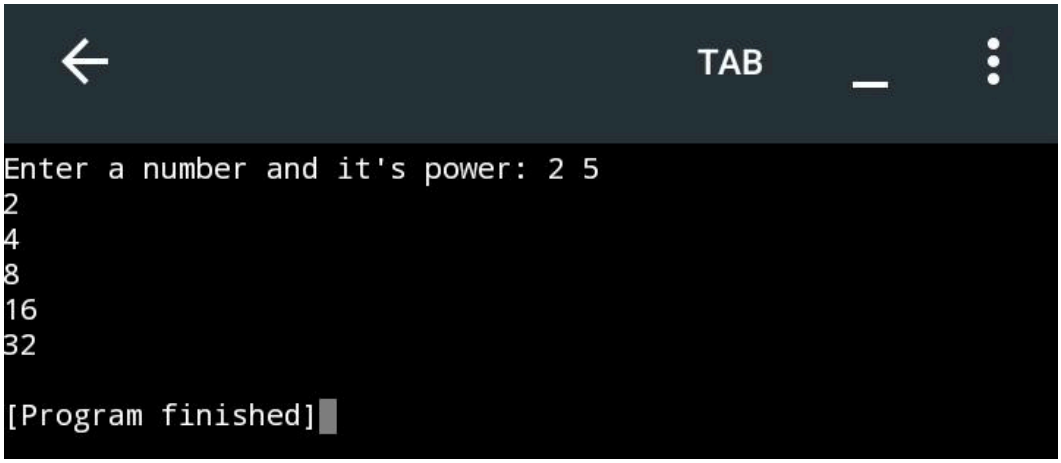
```
Enter a number: 2
2
4
6
8
10
12
14
16
18
20
```

Task – 02: (Find power of any number using for loop.)

Program:



Output:



TASK -02
(Find Power Of Number)

Program:

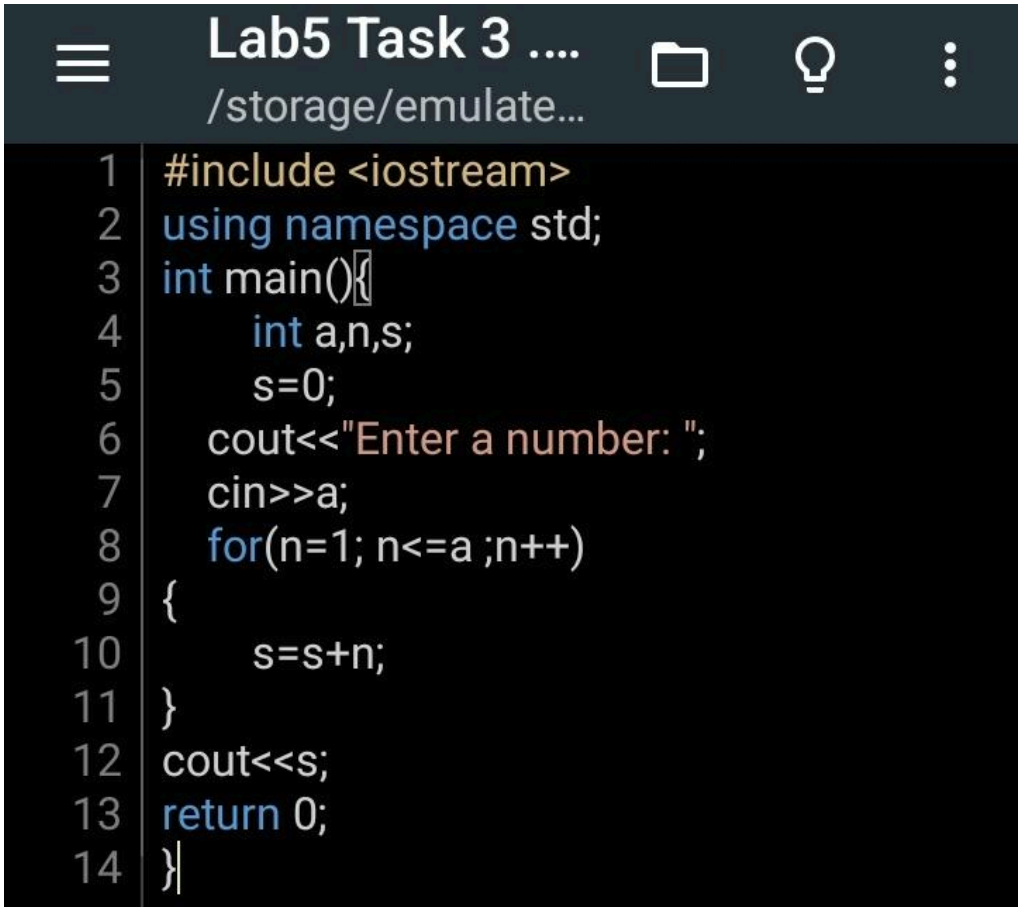
```
#include <iostream>
using namespace std;
int main(){
 int n, p, r =1;
 cout<<"Enter a number and it's power: ";
 cin >> n >> p;
 for (int i = 1; i<=p ; i++)
 {   r = r*n;
  cout<<r<<endl;
 }
return 0;
}
```

Output

```
Enter a number and it's power: 2 5
2
4
8
16
32
```

Task – 03: (Sum of Natural Numbers using loop.)

Program:



Output:



TASK - 03
(Sum Of Natural Numbers)

Program:

```
#include <iostream>
using namespace std;
int main(){
  int a, n, s;
   s = 0;
  cout << " Enter a number: ";
  cin >> a;
  for (n = 1; n <= a; n++)
  { s = s + n;
   }
   cout << s;
   return 0;
   }
```

Output

```
Enter a number: 5
15
```

Task – 04: (C++ program to find sum of digits of a number.)

Program:

Task 4 .cxx
/storage/emulate...

```
1  #include <iostream>
2  using namespace std;
3  int main(){
4     int n, a, s=0;
5  cout<<"Enter a three digit number: ";
6  cin>>n;
7  for( ; n > 0 ; n = n / 10 )
8  {
9   a = n % 10;
10  s = s + a;
11 }
12   cout<<s;
13    return 0;
14 }
```

Output:

TAB

```
Enter a three digit number: 358
16
[Program finished]
```

TASK - 04
(Find Sum Of Digit Of Num.)

Program:

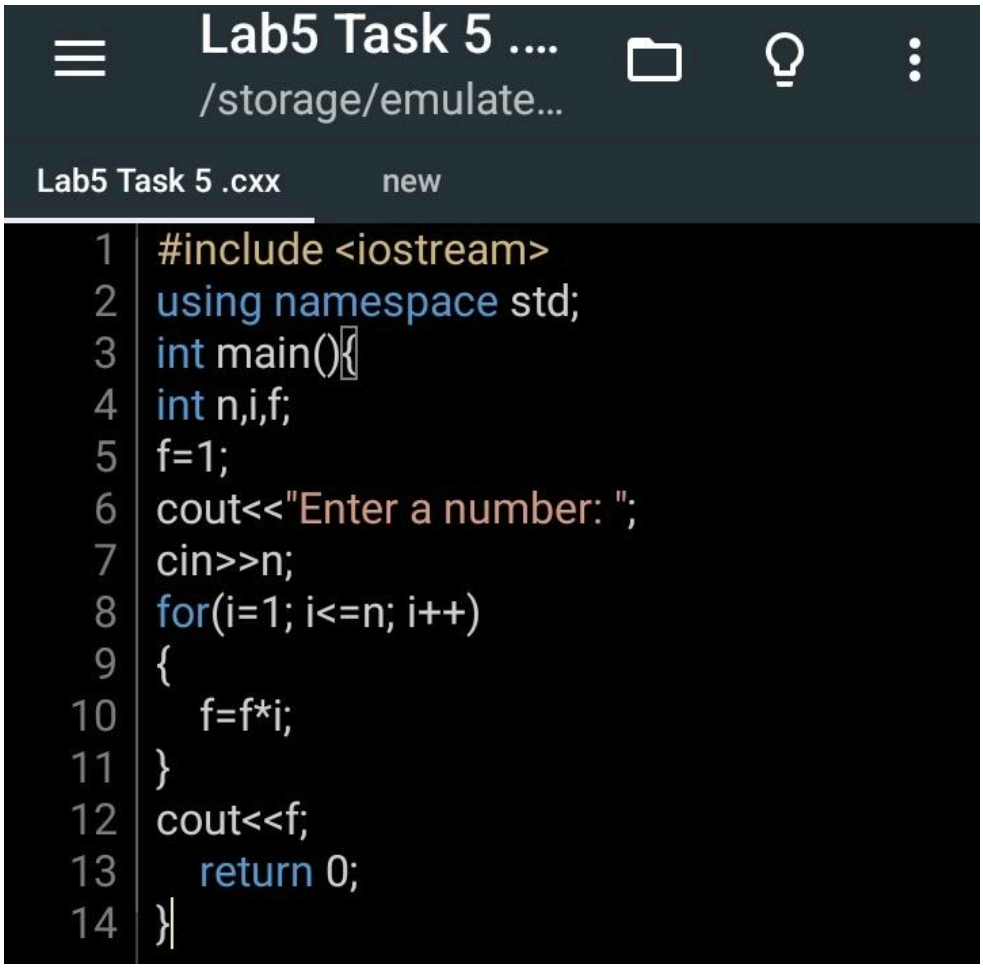
```
#include <iostream>
using namespace std;
int main(){
  int n, a, s=0;
 cout<<"Enter a three digit number: ";
 cin>>n;
 for( ; n>0 ; n=n/10)
 {  a= n%10;
     s= s+a;
  }
 cout<<s;
 return 0;
 }
```

Output

```
Enter a three digit number: 358
16
```

Task – 05: (Program to Generate Factorial a Certain Number.)

Program:



Output:



TASK - 05
(Factorial of A Number)

Program:

```
#include <iostream>
Using namespace std;
int main(){
 int n, i, f;
   f=1;
 cout<<"Enter a number: ";
 cin >> n;
 for( i=1; i<=n; i++)
 { f = f * i;
  }
  cout << f;
  return 0;
  }
```

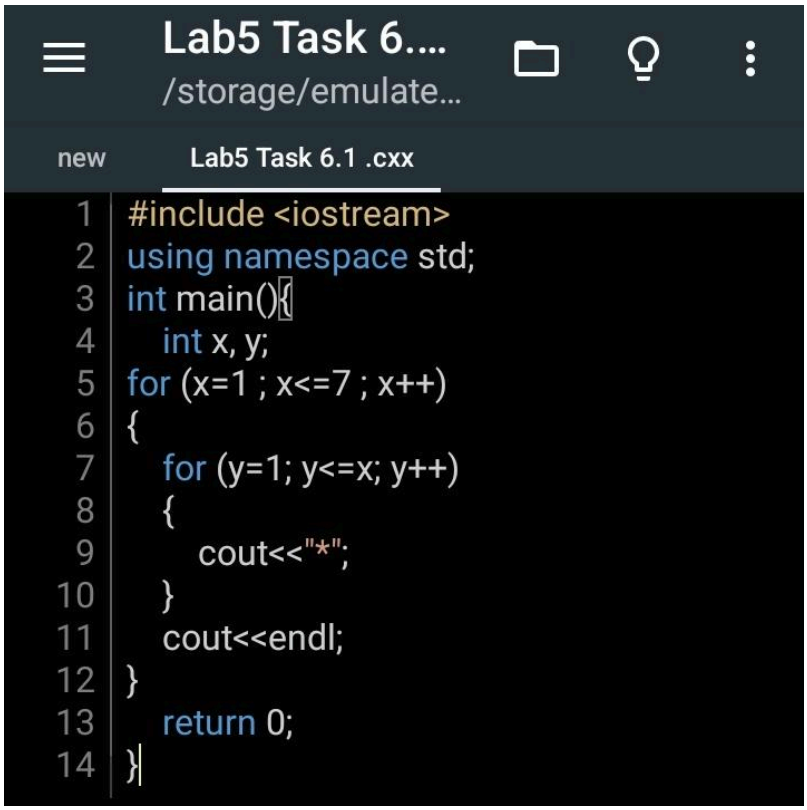
Output

```
Enter a number:  4
24
```

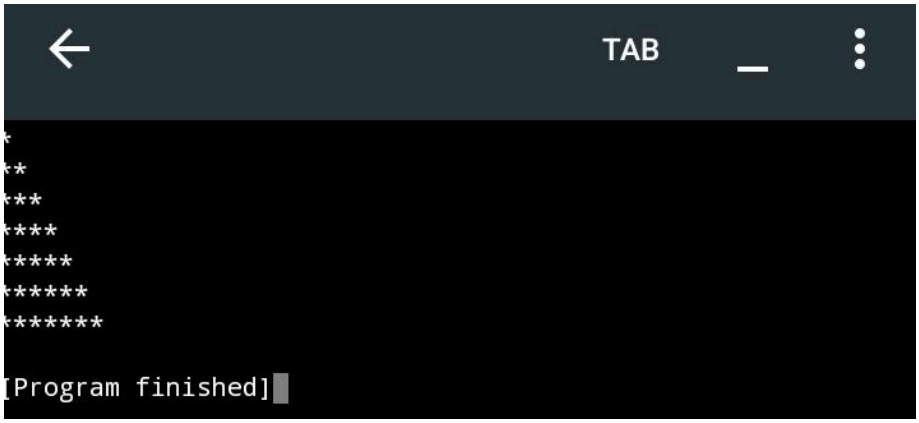
Task – 06: (Write a C++ program.)

A)Print the triangle of * using switch..

Program:



Output:



TASK - 06 (A)

(Print the following pattern.)

```
*
**
***
****
*****
******
*******
```

Program:

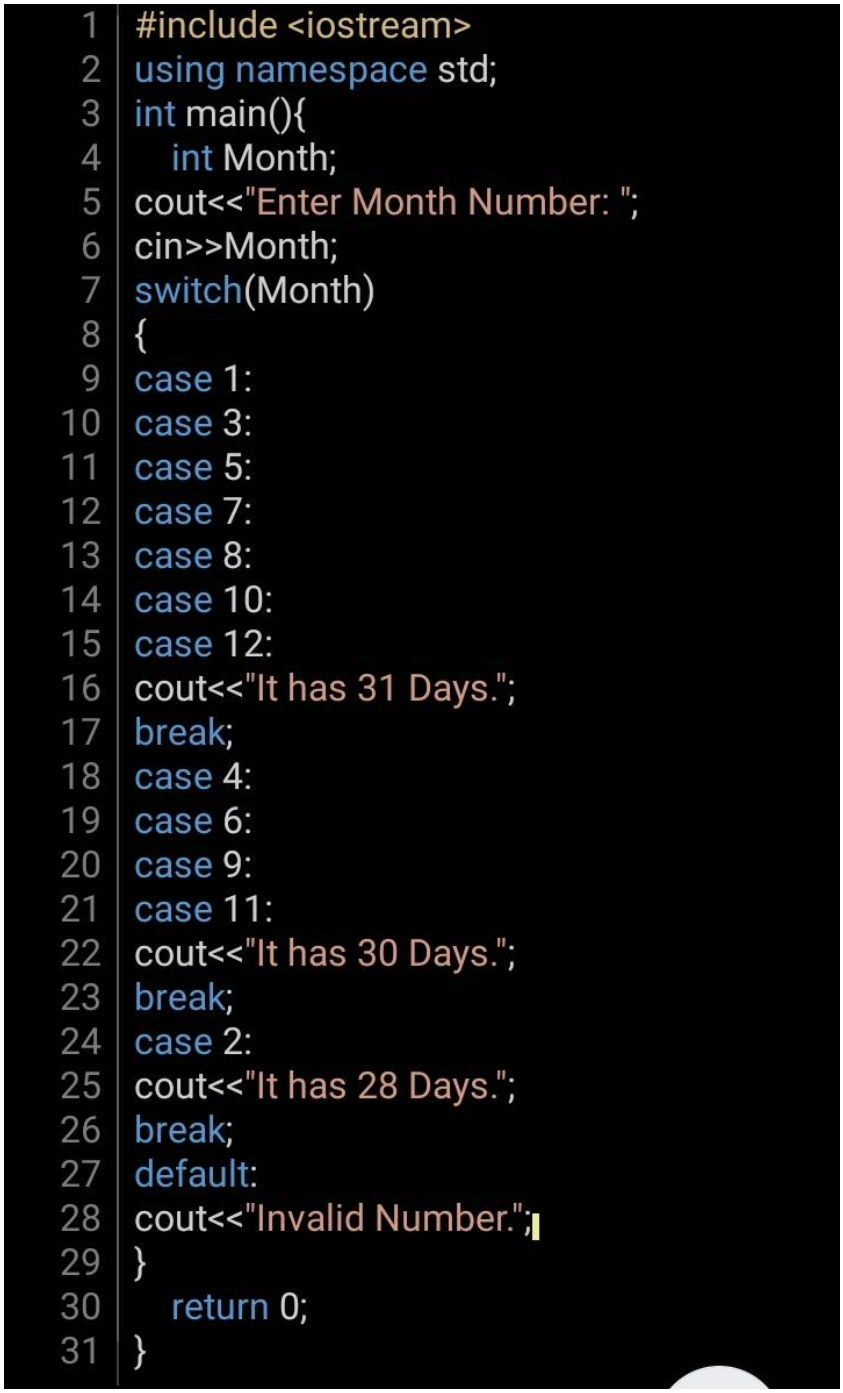
```
#include <iostream>
using namespace std;
int main(){
  int x, y;
for (x=1 ; x<=7 ; x++)
{
 for (y=1 ; y <=x; y++)
{ cout << "*" ; }
 cout << endl;
 }
 return 0;
 }
```

Output

```
*
**
***
****
*****
******
*******
```

B)Print total number of days in a month using switch case.

Program:



```
#include <iostream>
using namespace std;
int main(){
   int Month;
cout<<"Enter Month Number: ";
cin>>Month;
switch(Month)
{
case 1:
case 3:
case 5:
case 7:
case 8:
case 10:
case 12:
cout<<"It has 31 Days.";
break;
case 4:
case 6:
case 9:
case 11:
cout<<"It has 30 Days.";
break;
case 2:
cout<<"It has 28 Days.";
break;
default:
cout<<"Invalid Number.";
}
   return 0;
}
```

Output:

```
Enter Month Number: 7
It has 31 Days.
[Program finished]
```

TASK-06 (B)
(Total Number OF Days)

Program:

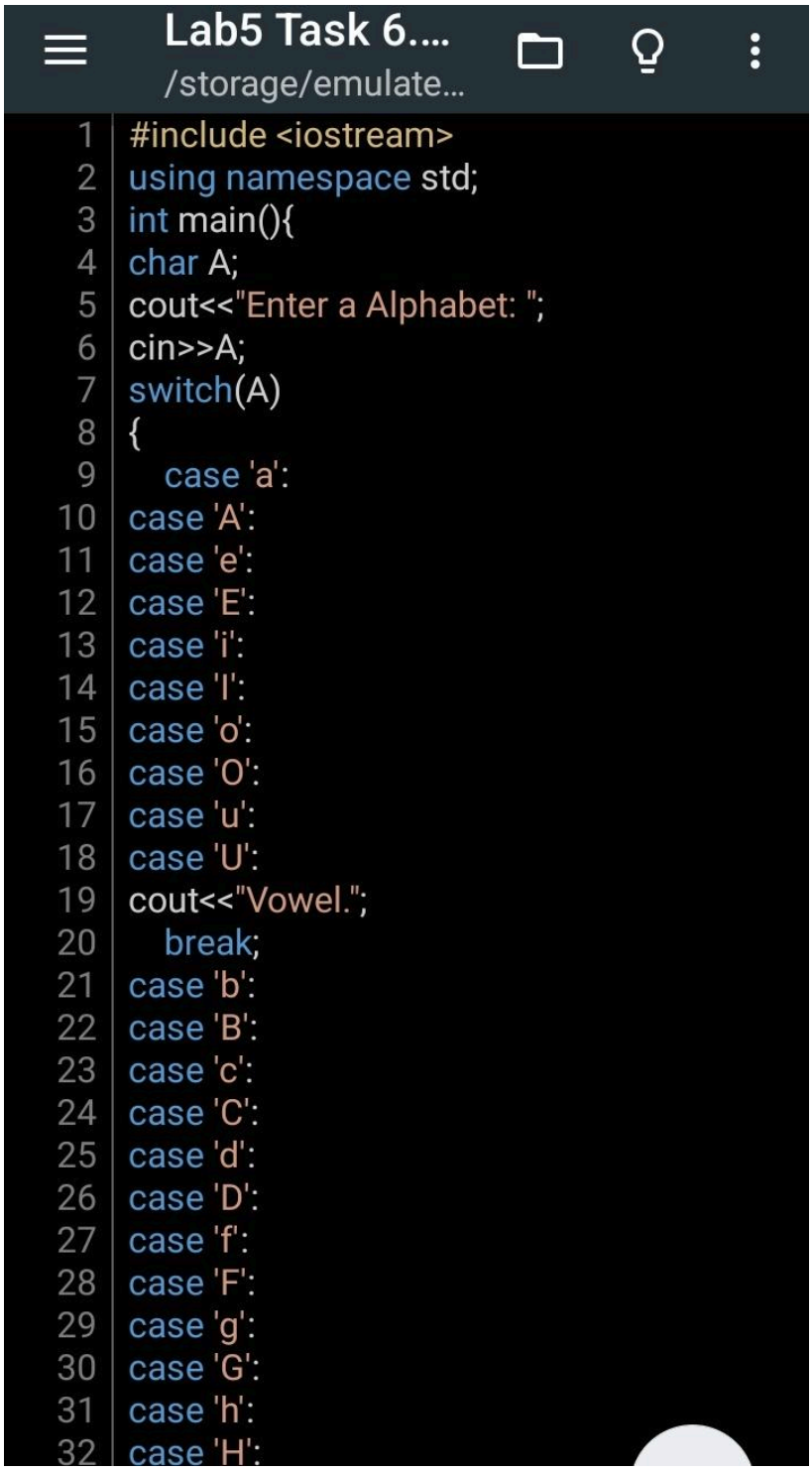
```
#include <iostream>
using namespace std;
int main(){
  int Month;
 cout<<"Enter Month Number: ";
 cin>>Month;
 switch(Month)
 { case 1:
   case 3:
   case 5:
   case 7:
   case 8:
   case 10:
   case 12:
  cout<<"It has 31 Days.";
   break;
    case 2:
  cout<<"It has 28 Days.";
   break;
    case 4:
    case 6:
    case 9:
    case 11:
  cout<<"It has 30 Days.";
    break;
   default;
   cout<<"Invalid Number.";
   }
   return 0;
   }
```

Output

```
Enter Month Number: 7
It has 31 Days.
```

C)Check whether an alphabet is vowel or consonant using switch case.

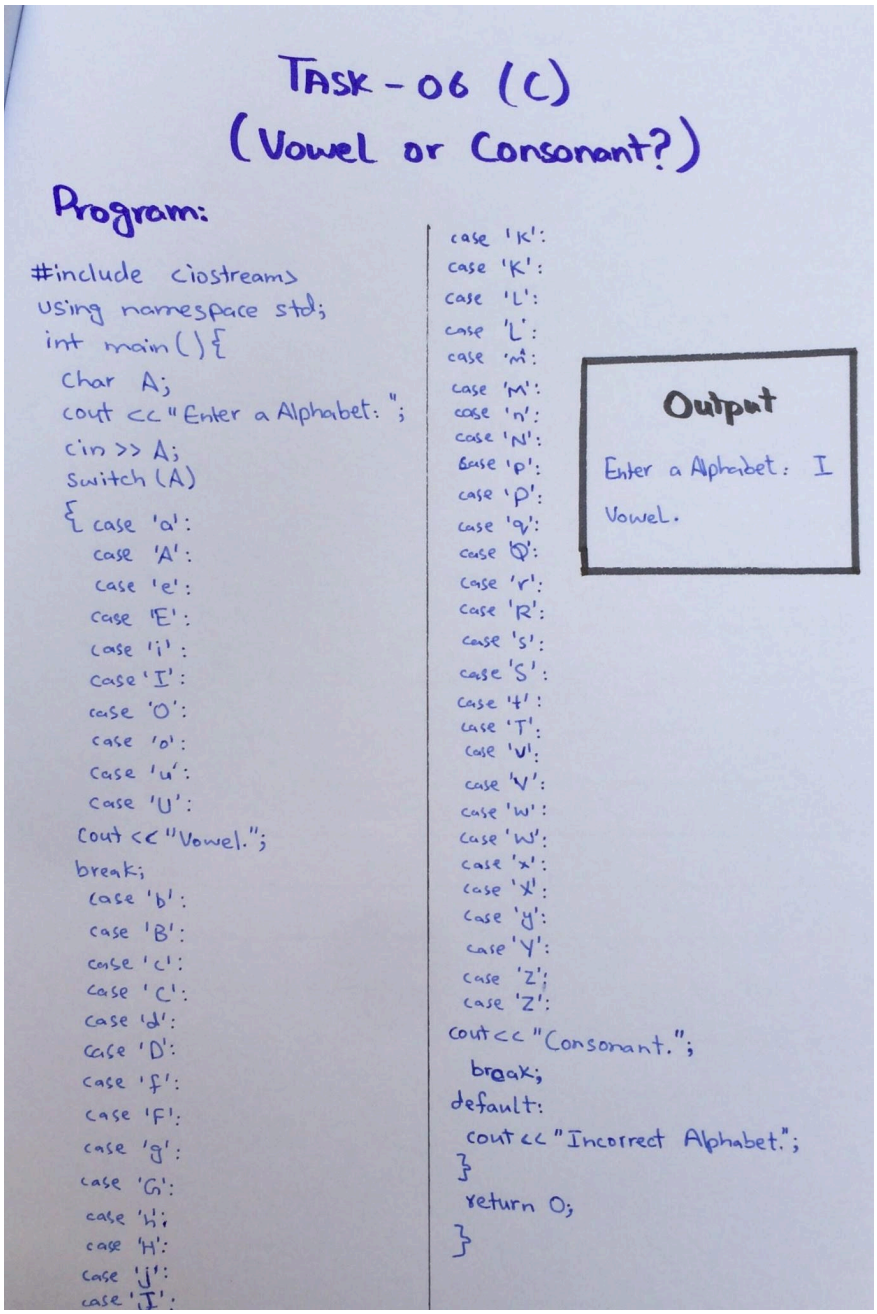
Program:



```
33  case 'j':
34  case 'J':
35  case 'k':
36  case 'K':
37  case 'l':
38  case 'L':
39  case 'm':
40  case 'M':
41  case 'n':
42  case 'N':
43  case 'p':
44  case 'P':
45  case 'q':
46  case 'Q':
47  case 'r':
48  case 'R':
49  case 's':
50  case 'S':
51  case 't':
52  case 'T':
53  case 'v':
54  case 'V':
55  case 'w':
56  case 'W':
57  case 'x':
58  case 'X':
59  case 'y':
60  case 'Y':
61  case 'z':
62  case 'Z':
63  cout<<"Consonant.";
64     break;
65  default:
66     cout<<"Incorrect Alphabet.";
67  }
68     return 0;
69  }
```

Output:





TASK - 06 (C)

(Vowel or Consonant?)

Program:

```
#include <iostream>
using namespace std;
int main(){
  Char A;
  cout << "Enter a Alphabet: ";
  cin >> A;
  Switch (A)
  { case 'a':
    case 'A':
    case 'e':
    case 'E':
    case 'i':
    case 'I':
    case 'O':
    case 'o':
    case 'u':
    case 'U':
   Cout << "Vowel.";
   break;
    case 'b':
    case 'B':
    case 'c':
    case 'C':
    case 'd':
    case 'D':
    case 'f':
    case 'F':
    case 'g':
    case 'G':
    case 'h':
    case 'H':
    case 'j':
    case 'J':
    case 'k':
    case 'K':
    case 'L':
    case 'l':
    case 'M':
    case 'm':
    case 'n':
    case 'N':
    case 'p':
    case 'P':
    case 'q':
    case 'Q':
    case 'r':
    case 'R':
    case 's':
    case 'S':
    case 't':
    case 'T':
    case 'v':
    case 'V':
    case 'w':
    case 'W':
    case 'x':
    case 'X':
    case 'y':
    case 'Y':
    case 'Z':
    case 'z':
cout<< "Consonant.";
  break;
default:
  cout << "Incorrect Alphabet.";
}
  return 0;
}
```

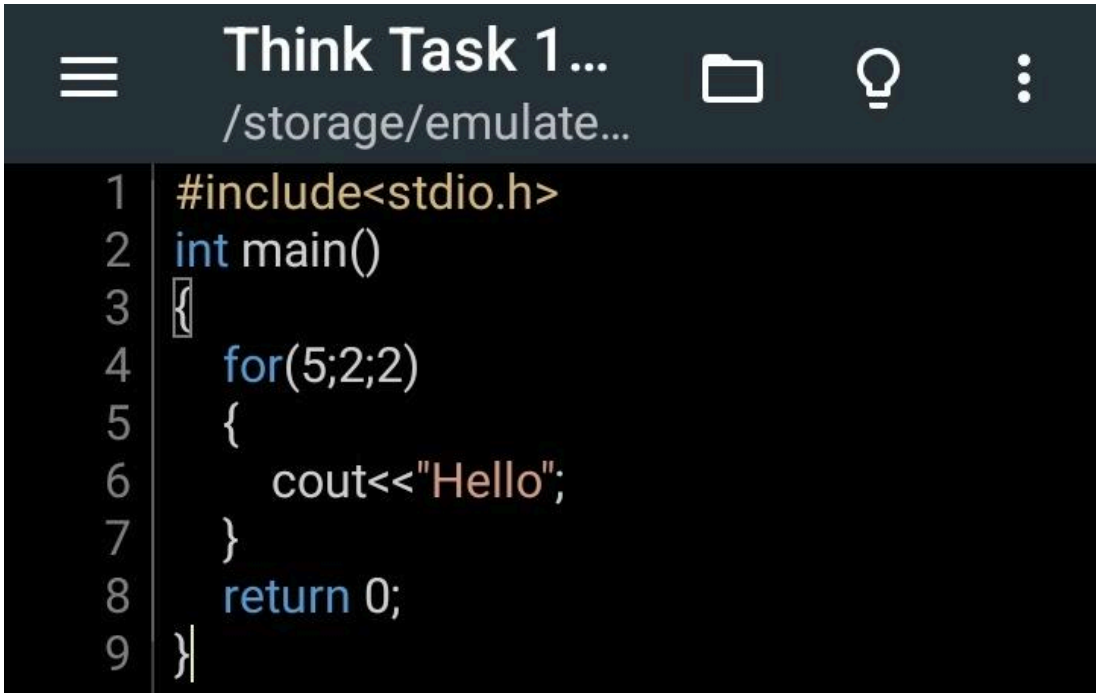
Output

Enter a Alphabet: I

Vowel.

THINK

Task – 01: (Observe the Code)

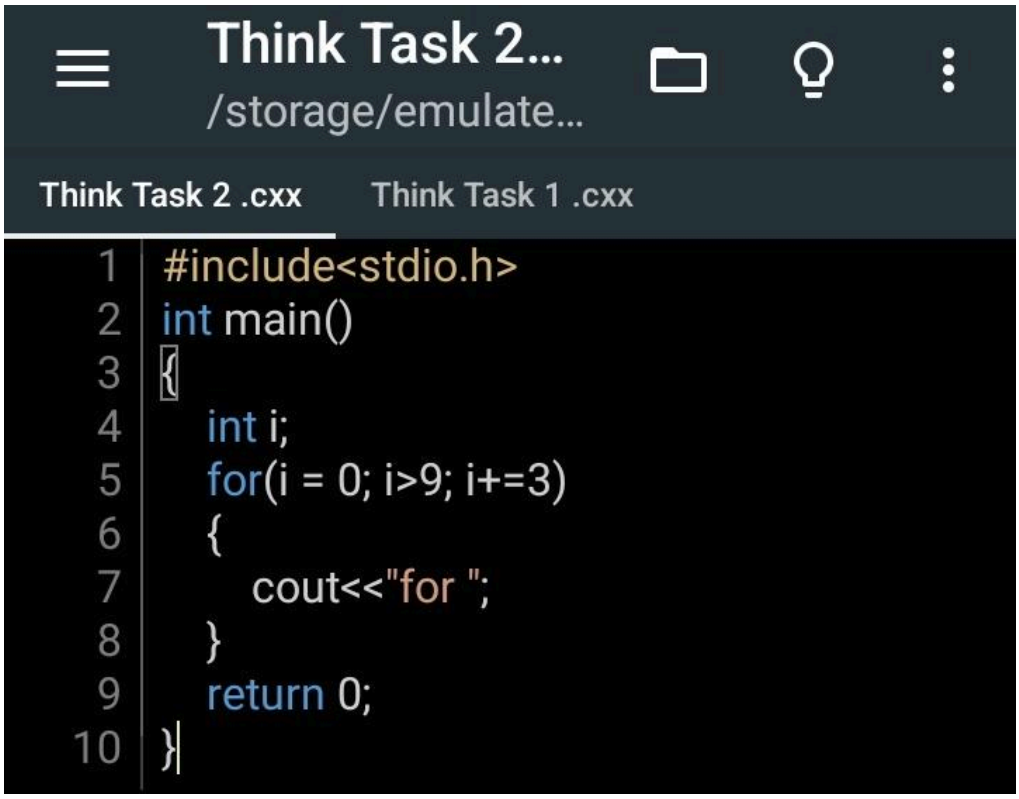


Output: Compilation Error

Reason: cout is C++ , not C. In C with #include<stdio.h> you can't use cout.

If you correct it to C++, then for(5;2;2) = infinite loop because condition 2 is always TRUE, so it will print Hello infinite times.

Task – 02: (Observe the Code)

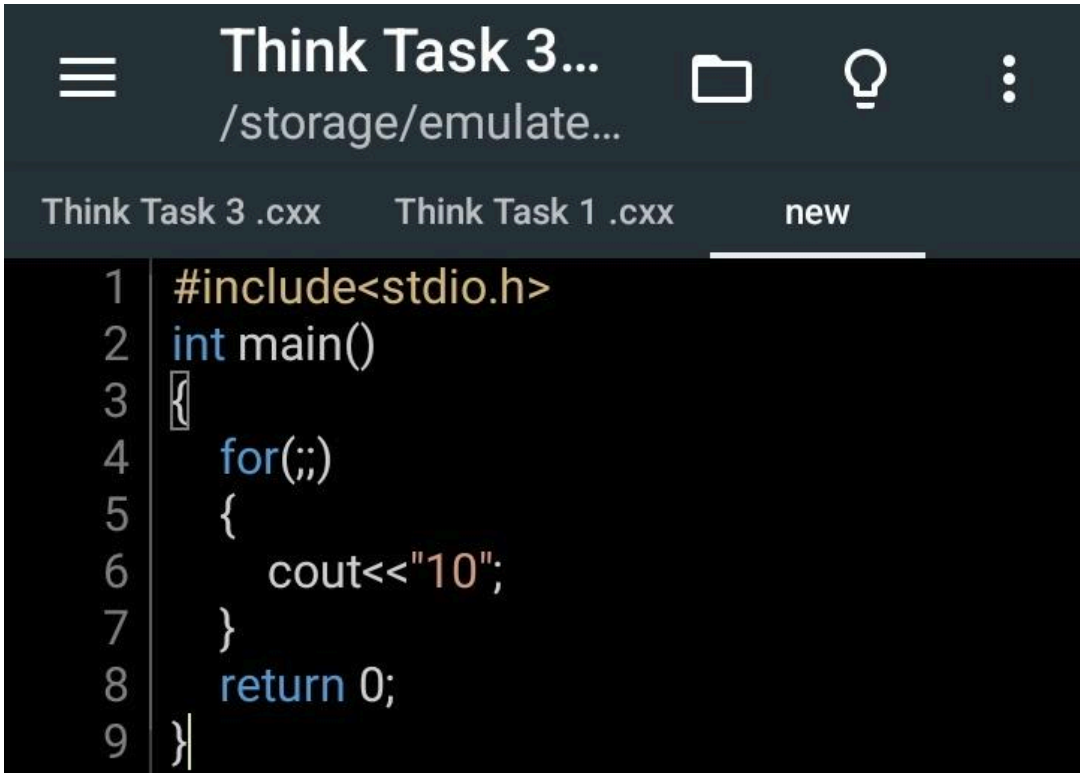


Output: Compilation Error

Reason: cout is C++ , not C. In C with #include<stdio.h> you can't use cout.

If you correct it to C++, then for(i=0; i>9; i+=3) = loop will not run even once because condition 0>9 is FALSE from the start, so it will print nothing / blank output.

Task -- 03: (Observe the Code)

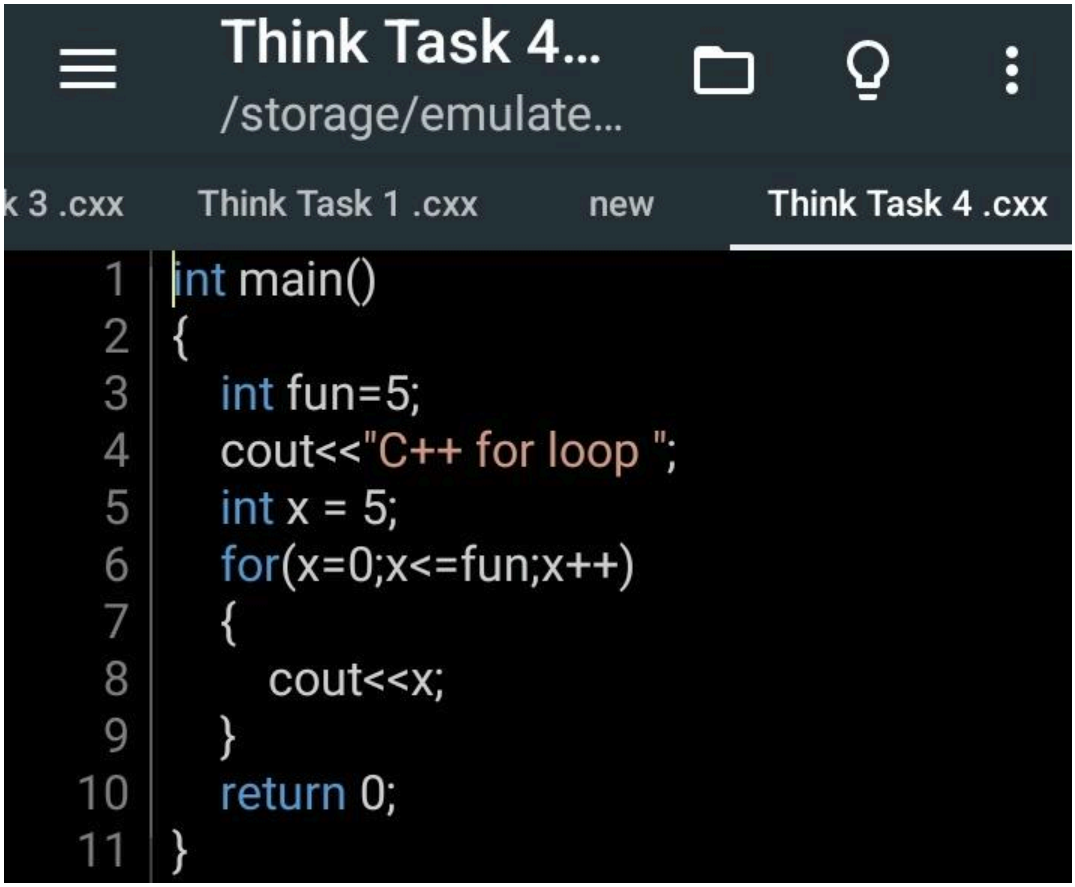


Output: Compilation Error

Reason: In this code cout is used but the header file is stdio.h. cout is part of C++ iostream library, so compiler will show error that cout is not declared. To use cout we need to include iostream and use std namespace.

If we correct the header, then for(;;) is an infinite loop because it has no condition to stop. So it will keep printing 10 continuously as 10101010 and never stop.

Task – 04: (Observe the Code)



Output: Compilation Error if run as given.

If corrected, output will be: C++ for loop 012345

Reason:

#include <iostream> and using namespace std; are missing so cout is not recognized by compiler.

After adding them, the loop for(x=0;x<=fun;x++) will run from 0 to 5 because fun=5, so it prints 0 1 2 3 4 5 after the text C++ for loop.